Nonisothermal and Isothermal Oxidation Behavior of Nb-Si-Mo Alloys

K. CHATTOPADHYAY, R. MITRA, and K.K. RAY

The nonisothermal and isothermal oxidation behavior of arc-melted, hypereutectic Nb-19Si-5Mo, Nb-18Si-26Mo, and hypoeutectic Nb-13Si-4Mo, and Nb-12Si-15Mo ternary alloys have been investigated *vis-à-vis* that of Nb and Nb-10Si alloy. The nonisothermal studies carried out using simultaneous thermogravimetric and differential thermal analysis in the temperature range of 50 °C to 1300 °C using heating rates of 5 °C, 15 °C/min, 25 °C/min, and 40 °C/min have shown oxidation initiation at lower temperatures for slower heating rates and higher Mo content. Isothermal oxidation experiments have been carried out in dry air at 800 °C, 1000 °C, and 1150 °C for 24 hours and the kinetics have been examined by the power-law equation. The studies show that the ternary Nb-Si-Mo alloys undergo less mass gain compared to the Nb and the Nb-10Si alloy. The hypo- or hypereutectic character of the alloys controls the oxidation behavior at 800 °C, while the effect of Mo content is predominant at temperatures of 1000 °C to 1150 °C. X-ray diffraction and scanning electron microscopy (SEM) accompanied by energy dispersive X-ray analysis have shown that the oxide scales of all the alloys are made of Nb_2O_5 and SiO_2. The beneficial effect of alloying with Mo is attributed to the improved sinterability of the oxide scale and increase in the activity of Si, which assists in the formation of a stable and continuous layer of SiO_2 at the alloy-oxide interface.

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I. INTRODUCTION

IN recent years, significant research has been directed toward the development of high-temperature materials for aerospace structural applications with operating temperatures exceeding that for most of the advanced Ni-base superalloys. The refractory Nb-Si–based alloys with high melting temperatures are of interest as candidate materials for aero-engine applications.[1–15] The microstructure of the Nb-Si–based alloys consists of hard and brittle intermetallic Nb_5Si_3 and relatively ductile Nb-solid solution (Nb_{ss}) phases. The Nb_5Si_3 exhibits excellent high-temperature strength retention and creep resistance,[1] but poor ductility and fracture toughness.[4] The resistance to cracking in these alloys (with microstructure showing a mixture of Nb_{ss} and Nb_5Si_3) is imparted by the ductile phase through intrinsic toughening mechanisms, such as crack arrest and bridging.[8,15] Therefore, an optimum volume fraction of Nb_{ss} and Nb_5Si_3 is required to achieve a desirable balance of mechanical properties involving room temperature fracture toughness, as well as high-temperature yield and creep strength, supplemented by appropriate oxidation resistance for high-temperature applications.

One of the major limitations of the Nb-based or Nb-Si–based alloys is their poor oxidation resistance at elevated temperatures. The molar volume of monoclinic β-Nb_2O_5 is 58.3 cm^3, while that of Nb is 10.9 cm^3,[16] suggesting that volume expansion on oxidation of Nb is large. Such a large volume expansion naturally causes high biaxial residual stress at the metal-oxide interface, leading to spallation. In addition, Nb_5Si_3 undergoes accelerated pest disintegration in the temperature range of 700 °C to 1000 °C, forming Nb_2O_5.[17] Complete disintegration of the Nb_5Si_3 has been observed on exposure at 1000 °C for 1 to 3 hours. The susceptibility of pest disintegration of Nb-silicides is commonly attributed to the mismatch in the coefficient of thermal expansion (CTE) of the oxidation products, along with the volume expansion accompanying the oxide growth. The CTE of Nb_2O_5 is one fourth that of Nb_5Si_3. The lower CTE of the oxide scale and the volume expansion during its formation leads to compressive residual stresses inside the scale and its spallation. The alloys with Nb_{ss}-Nb_5Si_3 mixture in the microstructure, thus suffer catastrophic oxidation upon exposure to air at temperatures above 500 °C, as the oxygen anions diffuse rapidly through the loose and porous oxide scales, which spall off easily.[18] Research efforts directed at improvement of oxidation resistance of the binary Nb-Si alloys have involved the addition of different alloying elements, such as Ti, Al, and Cr.[19–29] Ternary addition[2,26–29] of Al, Cr, or Ti in the Nb-Si alloy system improves the oxidation resistance, but degrades the creep resistance through decrease in the melting temperature. Previous studies by Chan[30,31] on cyclic oxidation behavior of the multiphase Nb-Si–based

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alloys have shown that the volume fractions of silicide, Laves, and Nb-solid solution phases strongly affect the change in mass in the temperature range of 900 °C to 1400 °C. The observations reported by Chan further suggest that: (a) the phases undergo oxidation independent of one another; (b) a larger volume fraction of Cr-rich Laves phase contributes to the formation of $CrNbO_4$ in the oxide scale, which is more protective than the other oxides; (c) a large volume fraction of Nb_{ss} is detrimental for oxidation resistance; and (d) it is almost impossible to achieve a continuous scale of SiO_2, which could have provided protection against elevated temperature oxidation.

Adding Mo to the Nb-based alloys is beneficial, as it raises the melting point and improves the high-temperature strength. Argent and Phelps[32] have reported that the Nb-Mo alloys (with Mo ≤ 1.9) oxidize more rapidly than pure Nb between 450 °C and 550 °C. In contrast, the alloys containing greater than 1.9 at. pct Mo have been found to oxidize at significantly lower rates, compared to the pure Nb at elevated temperatures. On exposure at 1000 °C for 1 hour, the binary Nb alloys containing 1 to 5 at. pct Mo have shown significantly less loss of mass compared to the pure Nb, while the Nb-20 at. pct Mo alloy has exhibited higher loss in mass.[33] On the other hand, alloying with Si leads to oxidation resistance worse than that of pure Nb.[33,34] Niobium alloyed with 1, 5, and 20 at. pct Si has shown greater damage due to oxidation on exposure at 1000 °C for 1 hour, compared to the pure Nb. Contrary to the observations of Miller and Cox,[33] Balasone *et al.*[34] have reported that alloying of the multicomponent Nb-Si–based alloys with a concentration of more than 1 at. pct Mo leads to severe degradation of the oxidation resistance. However, the existing information on the effect of alloying with Mo and its concentration on the oxidation behavior of the ternary Nb-Si-Mo alloys is limited to provide the required fundamental and in-depth understanding. The aim of the present work is to examine and compare the nonisothermal and isothermal oxidation behavior of the hypo-, hyper-, or near-eutectic, ternary cast Nb-Si-Mo alloys in the temperature range of 800 °C to 1150 °C in air, with emphasis on the kinetics of oxidation and constituents of oxide scales.

II. EXPERIMENTAL PROCEDURE

The raw materials used in the present study were Nb, Si, and Mo powders with purity of 99.7, 98.6, and 99.8 (wt pct), respectively. Four ternary and one binary alloy having an average chemical composition, Nb-19Si-5Mo (designated as alloy A), Nb-18Si-26Mo (alloy B), Nb-13Si-4Mo (alloy C), Nb-12Si-15Mo (alloy D), and Nb-10Si, were processed by arc melting of the elemental powder mixtures, containing appropriate weight fractions of Nb, Mo, and Si. The buttons were remelted at least 5 times to achieve chemical homogeneity. The arc-melted samples were cut by wire electrodischarge machining, for microstructural examination and oxidation studies. The as-cast specimens were examined using X-ray diffraction (XRD) on a PHILIPS* PW 1710 diffractometer to identify the phases present. The microstructures of the alloys were examined with the help of a JEOL** JSM 5800 scanning electron microscope (SEM) using secondary (SE) and back scattered electron (BSE) imaging modes, and the chemical compositions of the constituent phases were determined with the help of energy dispersive X-ray (EDX) analyses. The EDX analyzer (Oxford, United Kingdom) on the JSM 5800 SEM is equipped with a Si (Li) detector having an ultra-thin polymer window, which permits the detection of elements with a low atomic number starting from Boron. In addition, the volume fractions of the constituent phases in the as-cast specimens were determined from the recorded SEM images using point-counting method.

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Small specimens having mass less than 25 mg were subjected to simultaneous thermogravimetry (TG) and differential thermal analysis (DTA) on a Perkin-Elmer TG-DTA machine (Waltham, Massachusetts) to study the nonisothermal oxidation behavior. These experiments were carried out in air at the temperature range of 50 °C to 1300 °C with four different heating rates (5 °C, 15 °C, 25 °C, and 40 °C/min), using a Pt crucible and the inert Al_2O_3 powder as reference.

Rectangular bars having approximate dimensions of 10 mm × 4 mm × 3 mm were mechanically polished using diamond impregnated discs, and silicon carbide paper up to a grit size of 600 mesh for isothermal oxidation experiments. The polished specimens were wrapped in platinum wire, and then suspended inside the hot zone of a vertical tube furnace with the help of nichrome wire. The top end of the nichrome wire was hooked to the weighing pan of a mechanical balance, kept 1.5 m above the furnace opening on a steel support frame. The tip of a Pt-Pt/Rh thermocouple was put at the possible closest proximity to the sample for continuous monitoring of its temperature. The specimens were subjected to isothermal exposure at 800 °C, 1000 °C, and 1150 °C for time span of 24 hours. At least two specimens were examined for each condition of isothermal oxidation. The mass of the specimens was discretely recorded at intervals of 1 hour during the isothermal exposure. The constituent phases of the oxide scale were determined by XRD analysis. The microstructure and the chemical composition of the top surface of oxide scales were examined using SEM and EDX analyses, respectively. Subsequently, the oxidized alloys were mounted using a powder mixture of copper and bakellite, and metallographically polished to examine the scale cross section using the SEM and EDX analyzer.

III. RESULTS

The results of the microstructural examination of the arc-melted alloys for analysis of the volume fraction and composition of the constituent phases, as well as the studies of oxidation behavior, are described in this section.

A. *Microstructural Constituents*

The microstructures of the binary Nb-Si alloy and those of the ternary Nb-Si-Mo alloys (A, B, C, and D) in as cast condition, have been discussed in detail in earlier publications.[15,35] The alloys A and B with Si content, slightly exceeding that at the eutectic point (17.5 at. pct Si) in the binary Nb-Si phase diagram[36] are hypereutectic. On the other hand, the Nb-10Si alloy as well as the alloys C and D with Si concentration less than that at the eutectic point are hypoeutectic. Figures 1(a) and (b) show the typical SEM (BSE) images of the hypoeutectic Nb-10Si alloy and the ternary hypereutectic alloy B, respectively. The microstructure of Nb-10Si alloy comprises primary Nb_{ss} and eutectic mixture of secondary Nb_{ss} and Nb_5Si_3. The primary phase in the hypereutectic alloys, A and B is α-$(Nb,Mo)_5Si_3$, whereas the primary phase in the hypoeutectic alloys, C and D is Nb_{ss}. In all the ternary Nb-Si-Mo alloys, the eutectic mixture is made of Nb_{ss} and secondary α-$(Nb,Mo)_5Si_3$. Further, it may be noted that the Si content of the alloys A and B, and that of C and D, are almost identical, while the Mo content in each of these pairs is different. On the other hand, the alloys A and C possess almost similar concentration of Mo, but differ in their Si content. Table I shows the volume fractions of Nb_{ss} and α-$(Nb,Mo)_5Si_3$ determined by Rietveld analyses,[35] as well as the results of EDX analyses related to concentration of alloying elements present as substitutional solute in the Nb_{ss} and α-$(Nb,Mo)_5Si_3$ phases.[15,35] Because the oxidation resistance of the Nb_{ss} and $(Nb,Mo)_5Si_3$ phases are different, their volume fractions and compositions are important parameters in defining the oxidation behavior of the investigated alloys.

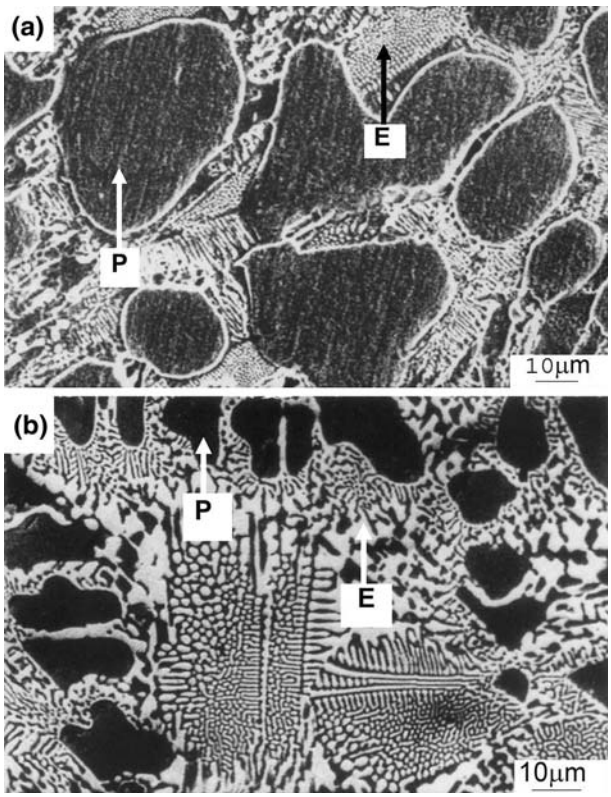


Fig. 1—Typical SEM (BSE) images of (*a*) the hypoeutectic Nb-10Si alloy and (*b*) the hypereutectic alloy B. The primary phase (Nb_{ss} in hypoeutectic and $(Nb,Mo)_5Si_3$ in hypereutectic) is marked as "P," and the eutectic is marked as "E".

B. *Oxidation Behavior*

The performance of the pure Nb, Nb-10Si alloy, and the ternary Nb-Si-Mo alloys, evaluated in terms of nonisothermal oxidation behavior on heating from room temperature to 1300 °C, isothermal oxidation kinetics in the range of 800 °C to 1150 °C, as well as the characteristics of the oxide scales are discussed.

1. *Nonisothermal oxidation*

The nonisothermal oxidation behavior of Nb, Nb-10Si, and the Mo-containing alloys A, B, C, and D are presented as TG and DTA plots in Figures 2, 3, 4. Figures 2(a) through (d) show the TG plots of change in mass against temperature, obtained at the four different heating rates for alloys A, B, C, and D, respectively. The plots representing the TG data of the Nb, Nb-10Si, and the Mo-containing ternary alloys, recorded at the heating rate of 5 °C/min, are depicted in Figure 3 for comparison. On examination of the plots in Figure 3, it is clear that the process of oxidation, characterized by detectable mass gain, is initiated at ≈525 °C in the alloys B and D with higher Mo content and at ≈700 °C in the alloys A and C with lower Mo concentration. The oxidation of Nb and Nb-10Si alloy also appears to initiate at ≈525 °C, and the net mass gain in these systems is higher, compared to that in the ternary Nb-Si-Mo alloys. At higher heating rates, the initiation of oxidation in all the alloys occurs at higher temperatures, as shown in Figures 2(a) through (d). Further, the TG plots in Figures 2(a) through (d) show that the net mass gain is generally higher at lower heating rates and *vice-versa*. This phenomenon can be conveniently explained with the argument that more time is available for oxidation at lower heating rates. However, a careful comparison of the TG plots (Figures 2(a) through (d)) shows that the difference between the temperatures of mass-gain initiation observed for the heating rates of 5 °C and 15 °C/min for the high Mo-containing alloys B and D is much greater than that in the case of alloys A and C with lower Mo content. Similar behavior is noted, if one compares the points of initiation of mass gain at heating rates of 5 °C/min and 25 °C/min. In other words, the influence of heating rate on the temperature for initiation of oxidation is more in the ternary alloys with higher Mo content than in those having a lower concentration. Figures 2(a) through (d) also indicate that the difference in mass-gain characteristics (point of

Table I. Volume Fractions and Solute Concentrations of Nb_{ss} and α-$(Nb,Mo)_5Si_3$ Phases in the Investigated Nb-Si-Mo Alloys Designated as A, B, C, and D[15]

	Nb_{ss}			α-$(Nb,Mo)_5Si_3$	
Alloy Designation	Vol Pct	Mo (At. Pct)	Si (At. Pct)	Vol Pct	Mo (At. Pct)
A (Nb-19Si-5Mo)	40.6	5.5 ± 0.4	2.24 ± 0.08	59.4	0.24 ± 0.07
B (Nb-18Si-26Mo)	44.8	22.3 ± 0.8	3.07 ± 0.03	55.2	4.2 ± 0.03
C (Nb-13Si-4Mo)	84.1	5.4 ± 0.5	2.6 ± 0.04	15.9	0.67 ± 0.05
D (Nb-12Si-15Mo)	88.0	20.5 ± 0.6	2.3 ± 0.06	12.0	4.7 ± 0.02

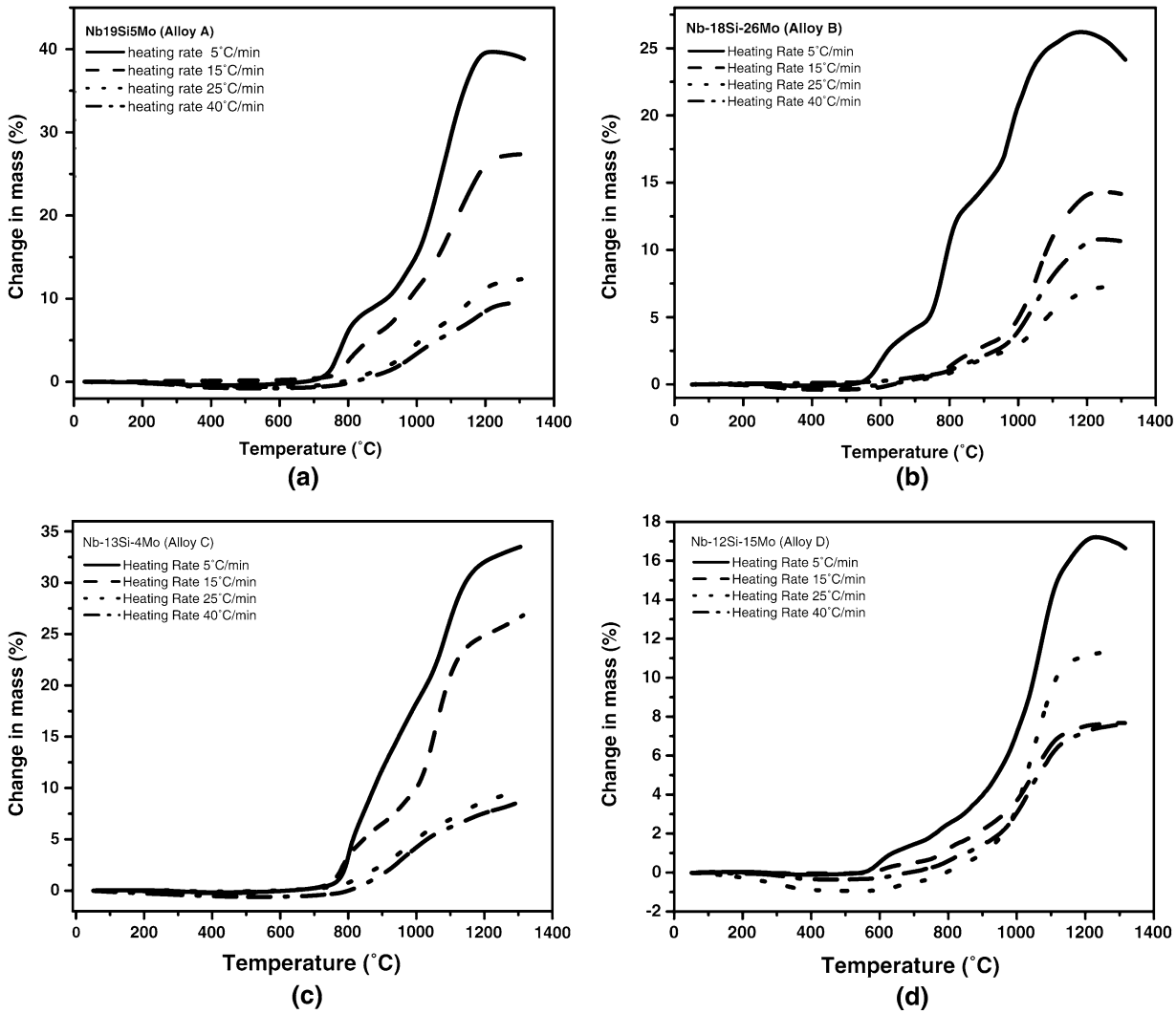


Fig. 2—Nonisothermal oxidation behavior depicted by the TG plots showing mass gain with temperature at the heating rates of 5 °C/min, 15 °C/min, 25 °C/min, and 40 °C/min for the alloys (*a*) A, (*b*) B, (*c*) C, and (*d*) D.

initiation and the net mass gain) corresponding to the heating rates of 25 °C/min and 40 °C/min is not very significant. Thus it may be inferred that the difference in the points of oxidation initiation and the total mass gains is reduced at higher heating rates.

The total mass gain as a fraction of the initial mass due to oxidation at the heating rate of 5 °C/min is the highest in the Nb-10Si alloy, followed by that of Nb, and then the alloys containing Mo (Figure 3). In addition, the mass gain due to oxidation for a given heating rate among the Mo-containing alloys is the highest in the case of alloy A, followed sequentially by that in the alloys C, B, and D. In other words, the alloys possessing a higher concentration of Mo have shown lower gain in mass during nonisothermal oxidation, regardless of the hypo- or hypereutectic character.

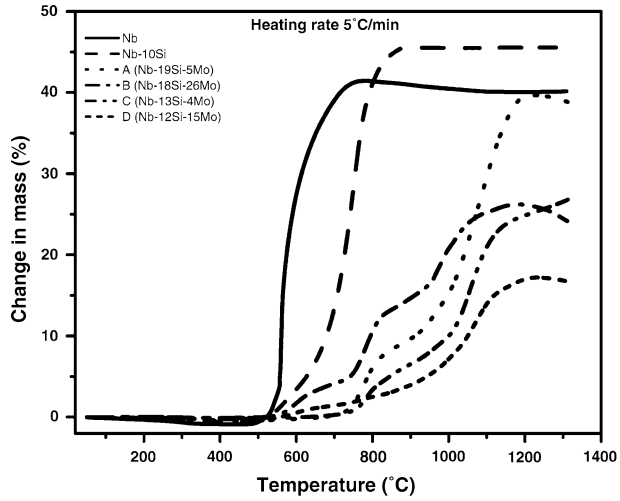


Fig. 3—TG plots of Nb, Nb-10Si alloy, and the alloys A, B, C, and D showing mass gain due to oxidation with increasing temperature at the heating rate of 5 °C/min.

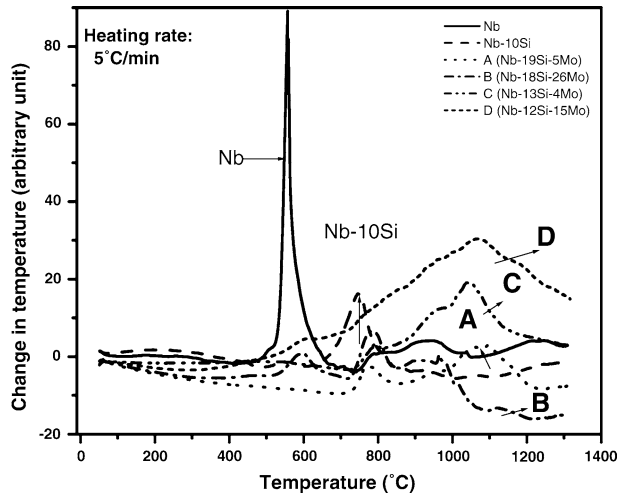


Fig. 4—DTA plots of the Nb, Nb-10Si alloy, and the alloys A, B, C, and D, showing the change in temperature on heating at the rate of 5 °C/min.

Interestingly, the temperature corresponding to the peak mass-gain varies marginally with heating rate, and is found to be in the range of 1170 °C to 1220 °C.

Figure 4 shows the DTA plots representing the oxidation behavior of Nb, Nb-10Si, and alloys A, B, C, and D for the heating rate of 5 °C/min. The first peak in the DTA plot for oxidation of pure Nb is observed at 557 °C, while that for oxidation of the Nb-10Si alloy is recorded at 538 °C. These results lead to the inference that oxidation of Nb_{ss} in the Nb-10Si alloy is initiated at 538 °C, and as a consequence the second peak for the Nb-10Si alloy at 747 °C should correspond to the start of oxidation of the Nb_5Si_3 phase. Using a similar logic, the peaks in the DTA plots corresponding to oxidation of Nb_{ss} and $(Nb,Mo)_5Si_3$ have been distinguished for the Mo-containing alloys and the results are compiled in Table II. The position of a characteristic DTA peak, representing oxidation of a specific phase varies from one alloy to another, probably because of the differences observed in the solute concentration. The results in Table I clearly show that the concentration of Mo in Nb_{ss} and $(Nb,Mo)_5Si_3$ varies significantly from one alloy to another, while that of Si hardly changes. Change in the Mo concentration is expected to alter the activity of the oxidizing species inside the Nb_{ss} and $(Nb,Mo)_5Si_3$, and hence the kinetics of their oxidation. The results in Table II are summarized as follows: (a) the temperature for initiation of oxidation for a particular phase is a function of its chemical composition; and (b) the lower Mo-containing alloys, A and C, exhibit a delayed start of oxidation compared to the ternary alloys having a significantly higher concentration of Mo.

Table II. Peak Positions of Nb, Nb-10Si, and Alloys A, B, C, and D in DTA Plots during the Exposure in the Temperatures Range of 50 °C to 1300 °C

Material	First Peak (Nb_{ss}) (T1), °C	Second Peak (Nb_5Si_3) or $(Nb,Mo)_5Si_3$ (T2), °C
Nb	557	—
Nb-10Si	538	747
Nb-19Si-5Mo	773	1096
Nb-18Si-26Mo	593	786
Nb-13Si-5Mo	790	1039
Nb-12Si-15Mo	604	1065

2. *Isothermal oxidation*

a. Oxidation kinetics. The oxidation kinetics at different temperatures has been studied in terms of the variation in mass gain per unit area with time. The results in Table III show the net mass gain observed in the Nb, Nb-10Si alloy, and alloys A, B, C, and, D on

Table III. Mass Gain per Unit Area in Nb, Nb-10Si, and Alloys A, B, C, and D during the Exposure at Different Temperatures for 1 Hour

Material	Temperature	Mass Gain per Unit Area after 1 Hour of Exposure (mg/cm^2)
Nb	800 °C	47 ± 06
	1000 °C	159 ± 15
Nb-10Si	800 °C	94 ± 13
	1000 °C	37 ± 4
	1150 °C	24 ± 3.8
A (Nb-19Si-5Mo)	800 °C	4 ± 1.5
	1000 °C	26 ± 5
	1150 °C	14 ± 2
B (Nb-18Si-26Mo)	800 °C	7 ± 1.4
	1000 °C	2 ± 1.4
	1150 °C	9 ± 1.7
C (Nb-13Si-4Mo)	800 °C	2 ± 1.3
	1000 °C	48 ± 8
	1150 °C	11 ± 2
D (Nb-12Si-15Mo)	800 °C	5 ± 1.6
	1000 °C	1 ± 1.2
	1150 °C	9 ± 2

1 hour of exposure at 800 °C, 1000 °C, and 1150 °C. The inferences derived from the results shown in Table III are summarized as follows.

(a) On oxidation at 800 °C, mass gain of the Nb-10Si alloy is twice that of the Nb. In contrast, the mass gain of Nb is more than 4 times that of the Nb-10Si alloy at 1000 °C.
(b) At 800 °C, the ternary alloys containing Mo exhibit mass gain of almost one order of magnitude less than those exhibited by the Nb and the Nb-10Si alloy.
(c) The mass gain at 1000 °C or 1150 °C is also less in the Mo-containing alloys, compared to that in the Nb and Nb-10Si alloy, although the difference is not as much as that observed for exposure at 800 °C.

Figures 5(a) through (c) show the plots displaying the isothermal oxidation kinetics of the alloys A, B, C, and D exposed for 24 hours at the temperatures of 800 °C, 1000 °C, and 1150 °C, respectively. The plots representing the oxidation kinetics at 800 °C (Figure 5(a)) show that the hypereutectic alloys A and B have almost similar rates of oxidation in the first 8 hours. The apparent sudden drops in the oxidation rate as well as the intermittent mass loss after 13 or 18 hours, observed in case of the alloy A, is suggestive of spallation. On the other hand, the alloy B does not show any obvious stable regime at 800 °C. In fact, the alloy B undergoes the highest mass gain amongst all the ternary alloys at 800 °C. In contrast, the hypoeutectic alloys C and D show a stable regime of almost no change in mass after just 2 hours. The oxidation kinetics at 800 °C certainly shows that irrespective of the Mo content, the hypereutectic Nb-Si-Mo alloys (having composition close to eutectic) are more prone to oxidation, compared to those having hypoeutectic composition.

Figure 5(b) shows that the oxidation kinetics displayed by the low Mo containing alloys A and C at 1000 °C are similar, irrespective of the fact that the former is hypereutectic and the latter is hypoeutectic.

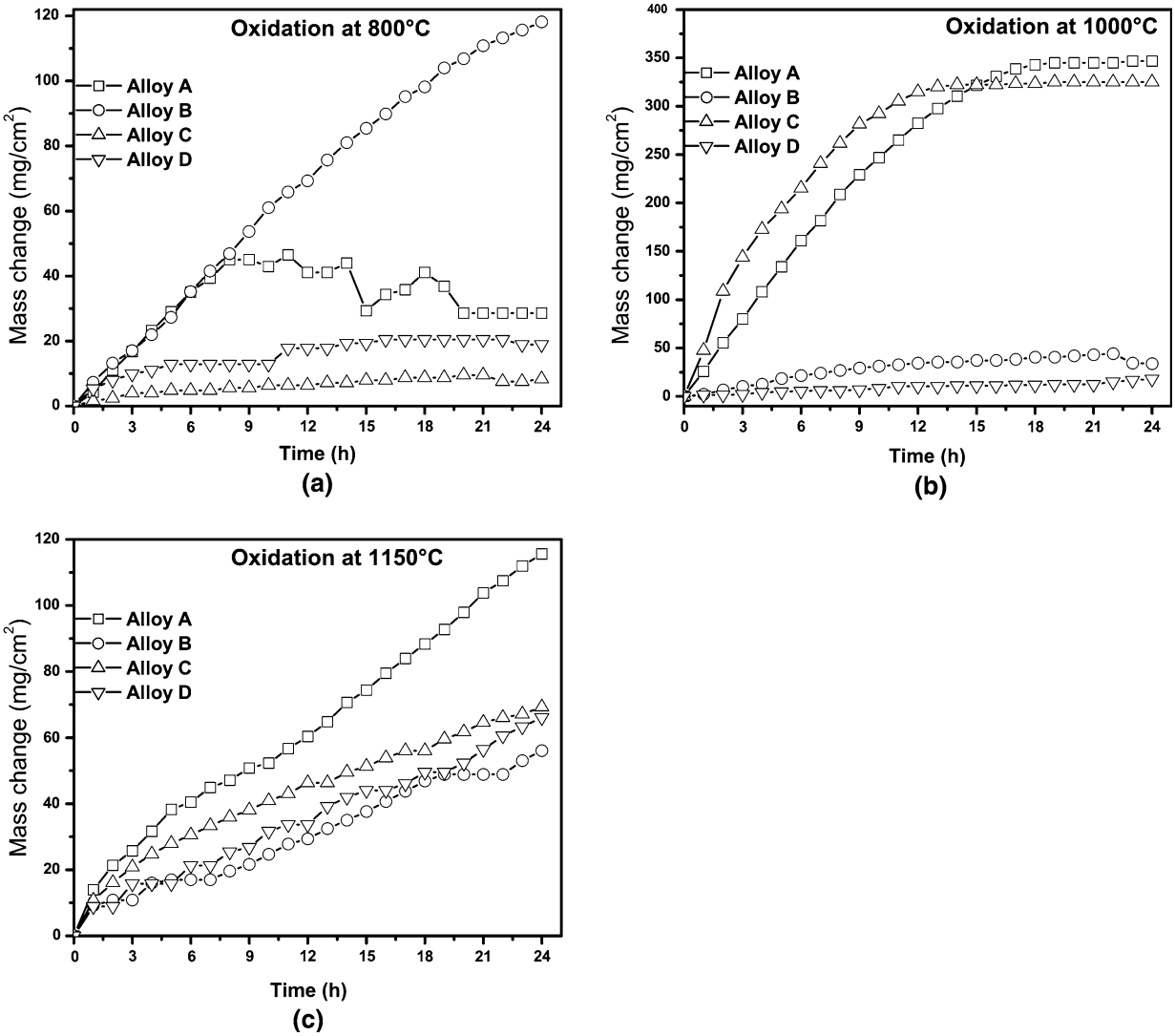


Fig. 5—Plots depicting the kinetics of oxidation of the alloys A, B, C, and D exposed for 24 h at (*a*) 800 °C, (*b*) 1000 °C, and (*c*) 1150 °C.

The rate of oxidation of alloy C is higher than that of alloy A up to about 17 hours, beyond which the trend reverses. After the initial accelerated rate of oxidation, the alloys A and C reach a stable regime at the 17th and 13th hour, respectively. The ability of the alloy C to reach a stable state of little or no change in mass earlier than the alloy A, is responsible for its lower net mass gain after 24 hours of exposure. On the other hand, the net mass gain in the high Mo containing, hypereutectic alloy B and hypoeutectic alloy D after 24 hours of exposure at 1000 °C is less by an order of magnitude, compared to those occurring in the alloys A and C. In fact, the alloy D appears to be distinctly the most resistant to oxidation both at 800 °C and 1000 °C, as shown in Figures 5(a) and (b). Thus examination of the oxidation kinetics at 1000 °C leads to the inference that the Nb-Si-Mo alloys with lower Mo content are more susceptible to oxidation at 1000 °C than those with higher concentration of Mo, regardless of the hypo- or hypereutectic composition.

In contrast to the stable regime of oxidation observed in case of exposure at 800 °C and 1000 °C, a continuous increase in mass along with absence of stable regime has been observed in all the Nb-Si-Mo alloys on exposure at 1150 °C, as depicted in Figure 5(c). However, the net mass gain after exposure for 24 hours at 1150 °C for each of the investigated alloys is relatively less, compared to that at 800 °C and 1000 °C. Again the ternary alloys, B and D with a larger concentration of Mo have shown much lower mass gain than the alloys A and C with lower Mo content. It is also interesting to note that the alloy B, in spite of its highest mass gain at 800 °C, has shown the least mass gain by oxidation at 1150 °C among all the investigated alloys after 24 hours of exposure. Hence, from comparison of oxidation kinetics at different temperatures, it is appropriate to infer that susceptibility of a given ternary Nb-Si-Mo alloy to oxidation is strongly dependent on the temperature of its isothermal exposure.

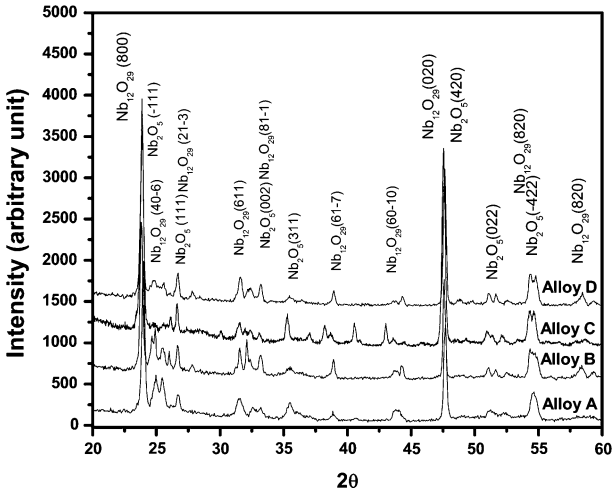


Fig. 6—Typical XRD patterns of oxide scale of the alloys A, B, C, and D, formed due to exposure at 1000 °C.

b. Oxide scale. The composition and morphology of the oxide scales are a function of temperature and alloy composition. Figure 6 depicts the XRD patterns obtained from oxide scales of all the investigated alloys showing the peaks representing β-Nb_2O_5 and $Nb_{12}O_{29}$, both having monoclinic structure. Furthermore, EDX analysis has shown distinct evidence for presence of a significant amount of SiO_2, besides the oxides of niobium. Absence of the peaks of crystalline SiO_2 in the XRD patterns implies that it is possibly present in amorphous form.

The morphology of the oxide scales appears to be closely related to the trends appearing in the kinetic data in Figures 5(a) through (c). The top surfaces and cross sections of the oxide scales, which correspond to severe mass gain or oxidation of the corresponding alloy at a given temperature, generally tend to be extremely rough with discontinuities, probably generated by spallation. Figure 7(a) shows the SEM (SE) image depicting the top surface of the oxide scale formed on the high Mo

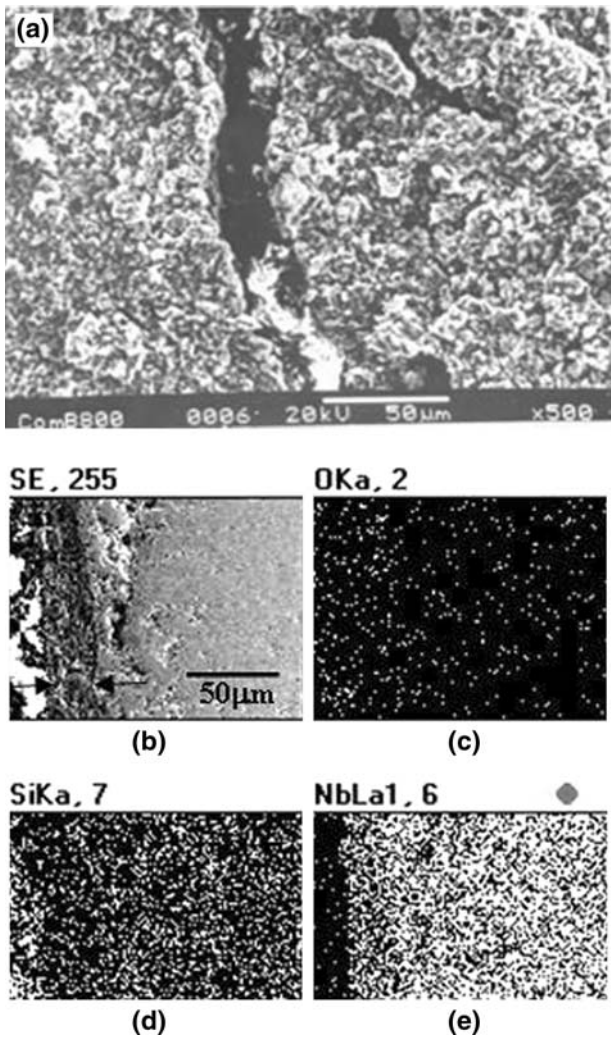


Fig. 7—The oxide scale formed on the alloy B exposed at 800 °C for 24 h: SEM (SE) images of the (*a*) top surface as well as (*b*) cross section, and EDX maps of (*c*) O, (*d*) Si, and (*e*) Nb. The oxide scale is shown with a pair of arrows in (b).

containing, hypereutectic alloy B due to exposure at 800 °C for 24 hours. The SEM (BSE) image of the oxide scale cross section and EDX maps of O, Si, and Nb are shown in Figures 7(b) through (e), respectively. The cracks are visible in the cross section beneath the oxide scale, as shown in Figure 7(b). The EDX maps (Figure 7(c) through (e)) confirm that the oxide scale has enrichment of both Nb and Si, suggesting that their oxides are present. The EDX point spectra from the randomly selected positions on the oxide scales have indicated a small concentration of Mo in the range of 2 to 4 at. pct, without any evidence of localized enrichment.

Figures 8(a) and (b) show SEM (SE) images depicting the top surface and cross section, respectively, of the oxide scale formed on the low Mo containing, hypereutectic alloy A due to exposure at 1000 °C for 24 hours. Cracks are visible in both the SEM images (Figure 8(a)

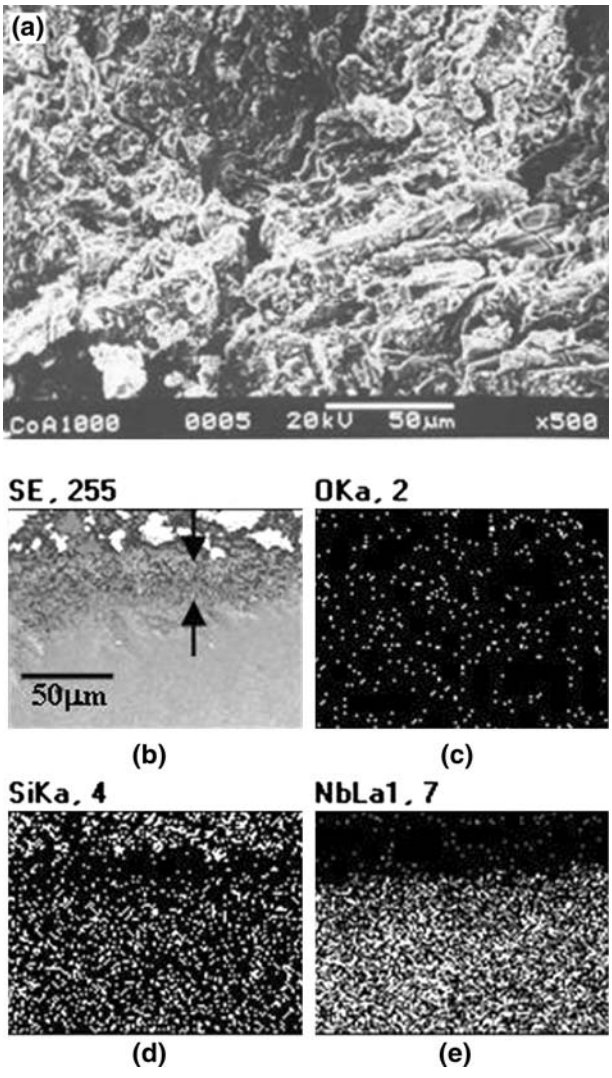


Fig. 8—The oxide scale formed on the alloy A exposed at 1000 °C for 24 h: SEM (SE) images of the (*a*) top surface, as well as (*b*) cross section, and EDX maps of (*c*) O, (*d*) Si, and (*e*) Nb. The oxide scale is shown with a pair of arrows in (b).

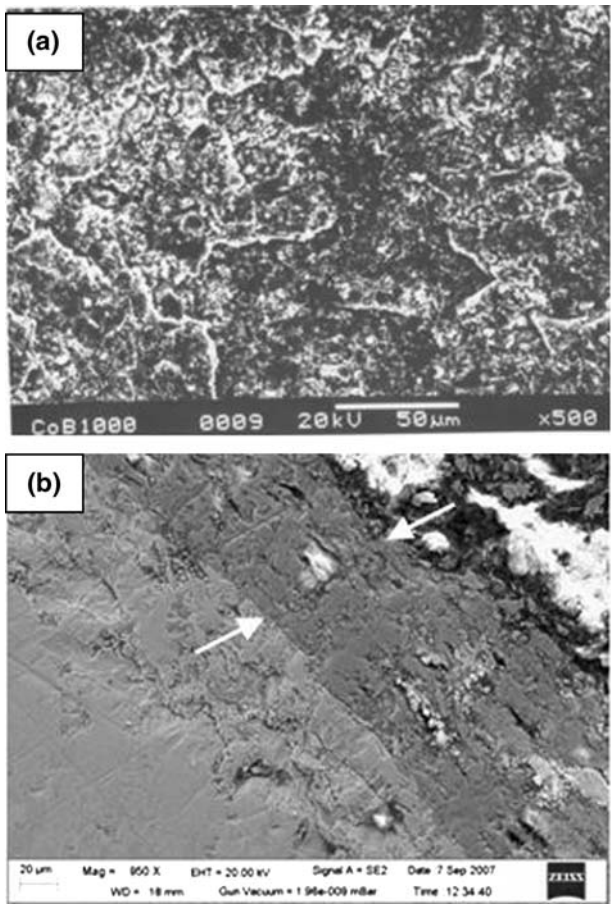


Fig. 9—SEM (SE) images of (*a*) the top surface and (*b*) the cross section of the oxide scale, formed on the alloy B on exposure at 1000 °C for 24 h. The oxide scale is shown with a pair of arrows in (b).

and (b)) of the oxide scale. Figures 8(c) through (e) depict the EDX elemental maps of Nb, Si, and O, respectively, indicating the presence of a mixture of the oxides in the scale. No trace of Mo could be found in the oxide scales. The top surface morphology and the cross section of the oxide scale on alloy B exposed at 1000 °C for 24 hours are shown in the SEM (SE) image of Figure 9(a) and the SEM (BSE) image of Figure 9(b), respectively. A few cracks or discontinuities can be observed inside the oxide scale presented in the SEM (BSE) image of Figure 9(b). However, the top surface (Figure 9(a)) of the oxide scale found on the alloy B after oxidation at 1000 °C, appears smoother than that formed on it on exposure at 800 °C (Figure 7(a)). In addition, EDX analyses have shown mixtures of Nb and Si oxides.

The top surfaces of oxide scales formed on the hypoeutectic and high Mo-containing alloy D at 800 ° or 1000 °C appears relatively smooth, compared to those of the other investigated alloys with low Mo content. Figures 10(a) and (b) show the SEM (SE) images depicting respectively, the top surface and cross section of the oxide scale formed on alloy D due to exposure at 1000 °C for 24 hours. Figures 10(c) through

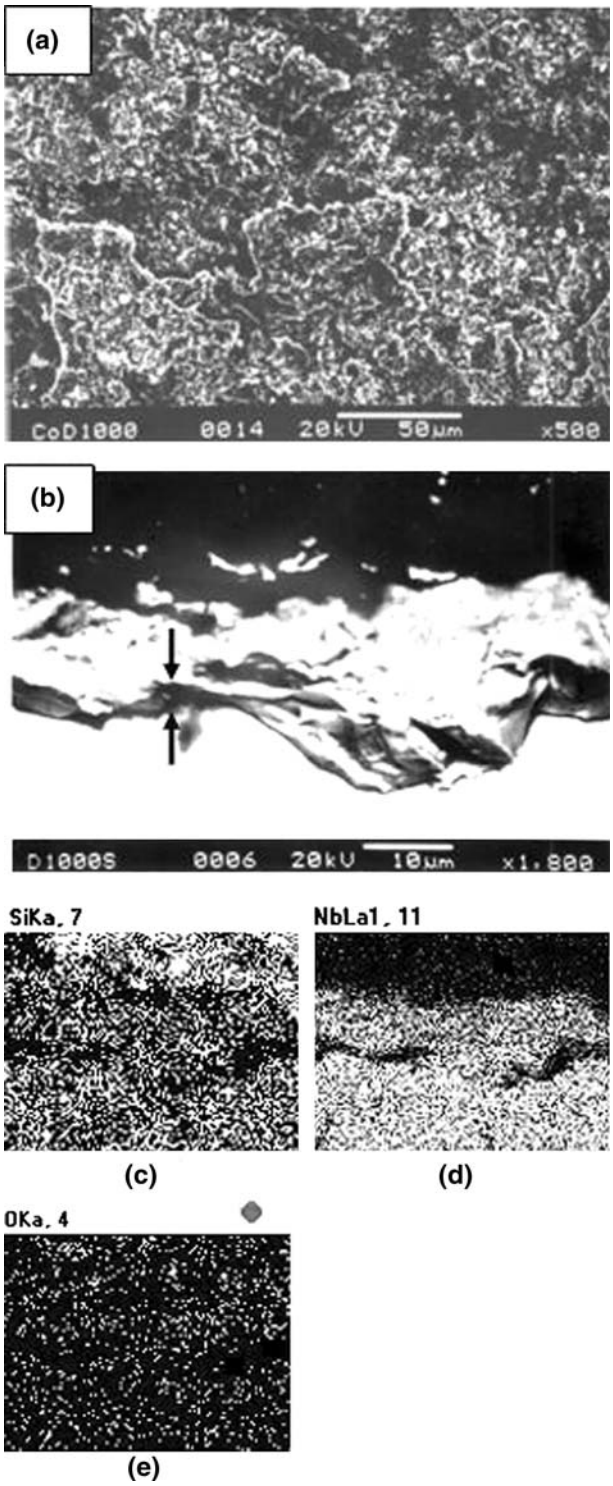


Fig. 10—Oxide scale formed on the alloy D exposed at 1000 °C for 24 h: SEM (SE) images of the (*a*) top surface, as well as (*b*) cross section, and EDX maps of (*c*) Si, (*d*) Nb, and (*e*) O. The SiO_2 layer at the metal-oxide interface is shown with a pair of arrows.

(e) show the EDX maps of Si, Nb, and O, respectively. It is also obvious from Figure 10(b) that a nonuniform layer of SiO_2 is formed at the alloy-oxide scale interface.

The top surfaces of the oxide scales formed during exposure at 1150 °C show varying degree of roughness.Figures 11(a) through (d) show the morphology of the top surfaces of oxide scales formed on the alloys A, B, C, and D, respectively, due to their exposure at 1150 °C. A series of cracks is present in most of the oxide scales. However, a qualitative comparison of the top surfaces of the oxide scales formed on the alloys A and C due to exposure at 1150 °C (Figures 11(a) and (c)) with those formed at 1000 °C (as shown typically in Figure 8(a)) indicates that the roughness is relatively less in the oxide scales formed at the higher temperature. Typical cross-sectional image and line scans representing the Nb and Si enrichments of the oxide scale formed on the alloy B during exposure at 1150 °C are shown in Figures 12(a) and (b), respectively. The oxide scale of the alloy B consists of a SiO_2 layer of 40- to 50-μm thickness at the alloy-oxide interface, followed by an outer thick layer of Nb_2O_5. Similar to the structure of its oxide scale formed at 1000 °C, the alloy D has also shown an approximately 50-μm-thick layer of SiO_2 at the alloy-oxide interface on exposure at 1150 °C, as depicted in Figure 13 and confirmed through EDX analyses. The outer layer made of Nb oxides has a thickness of several hundred micrometers, and shows discontinuous transverse cracks. On the other hand, no cracks are observed at the interface between the alloy and the SiO_2 layer, suggesting that the adherence is excellent. Analysis using EDX line scans across the oxide scales of the alloys B and D subjected to exposure at 1150 °C, have shown distinct evidence of oxygen penetration to a depth of about 50 μm beneath the SiO_2 layer. However, the SEM (BSE) images of the oxide scale cross sections (Figures 12(a) and 13) do not show any evidence of oxidation beneath the SiO_2 layer, implying that oxygen is present mainly as interstitial solute, and its activity is not high enough to cause oxidation.

IV. DISCUSSION

Nonisothermal and isothermal thermogravimetric studies have shown that the Nb-Si-Mo alloys are susceptible to oxidation to varying degrees at different temperatures, depending upon the composition. The oxidation mechanism in the investigated alloys can be understood through a detailed analysis of the kinetic parameters and the reactions leading to the formation of the oxides.

A. *Oxidation Reactions*

The probable oxidation products are expected to be the oxides of the constituent elements present in the investigated alloys. The free energy of formation[37] per mole of oxygen is the least for SiO_2, followed by that of Nb_2O_5 and MoO_2 as shown in Table IV. Amongst the possible Nb oxides, Nb_2O_5 does not possess the lowest free energy of formation per mole of oxygen. However, the published literature on the oxidation of either pure Nb or the Nb-based

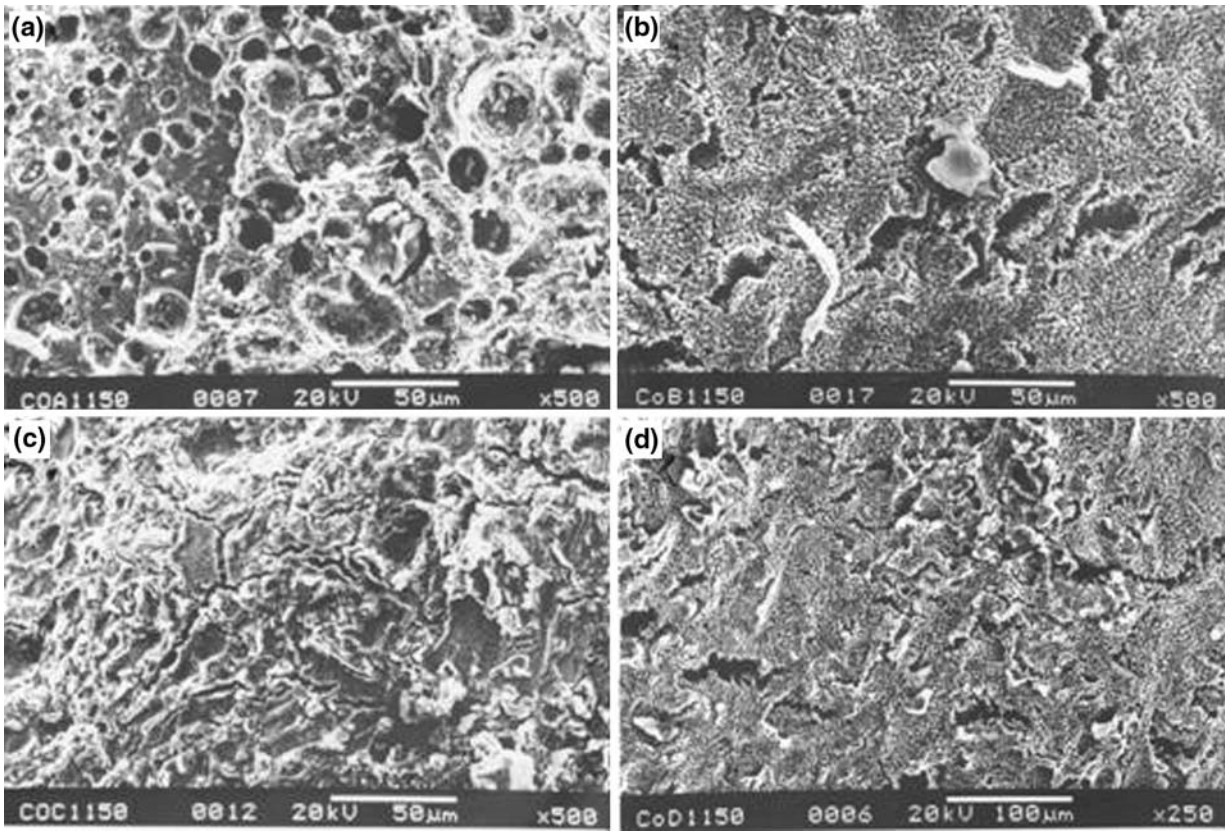


Fig. 11—SEM (SE) images depicting the morphology of the top surface of the oxide scales formed on the alloys A, B, C, and D due to the exposure at 1150 °C.

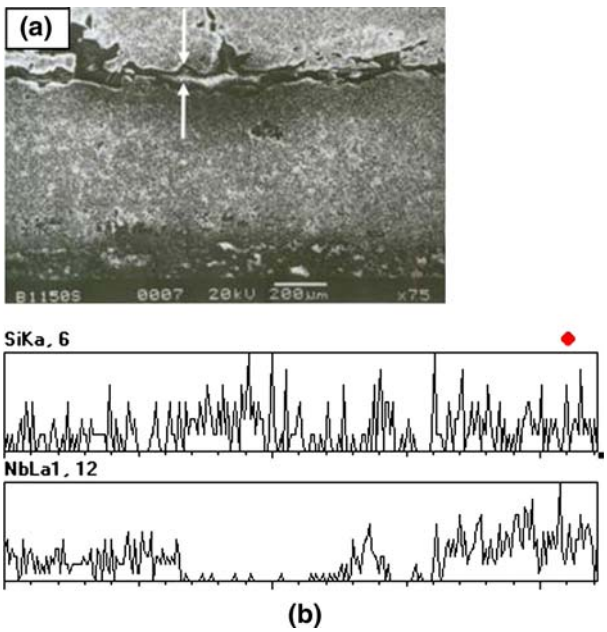


Fig. 12—Oxide scale formed on the alloy B exposed at 1150 °C: (*a*) SEM (BSE) image of the cross section and (*b*) the line scans of Si and Nb, respectively. The SiO_2 layer at the metal oxide interface is shown with a pair of arrows in (a).

alloys, reports the presence of Nb_2O_5 in the oxide scales.[16–18,22,24,30–34] Thus the formation of Nb_2O_5 in the oxide scale is kinetically driven. The possible

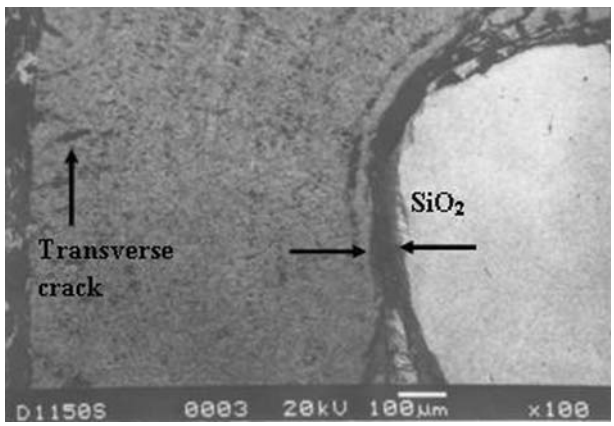


Fig. 13—SEM (BSE) image of the cross section of the oxide scale formed on the alloy D due to the exposure at 1150 °C for 24 h. The SiO_2 layer at the metal oxide interface is shown with a pair of arrows. Transverse cracks (single arrow) are visible in the oxide scale.

oxidation reactions involving the metallic Nb or the Nb_{ss} phases are

$$(4/5)Nb + O_2 = (2/5)Nb_2O_5 \quad [1]$$

$$(24/29)Nb + O_2 = (2/29)Nb_{12}O_{29} \quad [2]$$

$Nb_{12}O_{29}$ appears to be an intermediate reaction product of Nb and oxygen, caused by oxygen deficiency. In

Table IV. Standard Free Energies and Activities of Nb and Si Required for the Formation of Nb_2O_5, SiO_2, and MoO_2

Thermodynamic Parameter	800 °C	1000 °C	1150 °C
$\Delta G_1 = 0.4\Delta G_f(Nb_2O_5)$ (kJ/mol)	−572.6	−539.5	−514.9
K_1	7.53×10^{27}	1.37×10^{22}	7.97×10^{18}
a_{Nb}	1.0×10^{-34}	1.49×10^{-27}	1.66×10^{-23}
$\Delta G_2 = \Delta G_f(SiO_2)$ (kJ/mol)	−717	−683.16	−657.57
K_2	8.04×10^{34}	1.08×10^{28}	1.38×10^{24}
a_{Si}	5.92×10^{-35}	4.41×10^{-28}	3.46×10^{-24}
$\Delta G_3 = \Delta G_f(MoO_2)$ (kJ/mol)	−383.3	−365.8	−330.9
K_3	4.57×10^{18}	1.02×10^{15}	1.40×10^{12}
a_{Mo}	1.04×10^{-18}	4.65×10^{-15}	3.39×10^{-12}

addition, the atomic Si present in the Nb_{ss} and diffusing through the alloy would oxidize through the reaction

$$Si + O_2 = SiO_2 \quad [3]$$

The formation of Nb_2O_5 and SiO_2 by oxidation of Nb_5Si_3 phase is also possible through the reaction

$$(4/37)Nb_5Si_3 + O_2 = (10/37)Nb_2O_5 + (12/37)SiO_2 \quad [4]$$

The possible oxidation reactions involving Mo are

$$2Mo + 3O_2 \rightarrow 2MoO_3 \quad [5]$$

$$Mo + O_2 \rightarrow MoO_2 \quad [6]$$

MoO_3 undergoes sublimation at 704 °C, and hence its presence is not expected after an extended isothermal exposure at 800 °C or above. In fact, the MoO_3 starts vaporizing above 500 °C, and its vapor pressure rises sharply from 10^{-2} Pa to 10^{2} Pa, as the temperature is increased to 700 °C.[38] Hence, a more stable oxide of Mo, MoO_2 has been observed in the oxide scales of the Mo-Si-B alloys at temperatures exceeding 800 °C,[39,40] and is preferred to the MoO_3 in the present study as well. The free energies of formation of Nb_2O_5 and SiO_2 per mole of oxygen are much less than those of MoO_3 and MoO_2.[3,37]

The activity of each element in the investigated alloys also plays a significant role in controlling the formation of oxide. The oxidation reactions of Nb, Si, and Mo as depicted earlier by Eqs. [1], [3], and [6] are associated with free energy of formations of the corresponding oxides in the following way:

$$\Delta G_1 = -RT\ln K_1 = -RT\ln\frac{(a_{Nb_2O_5})^{0.4}}{(a_{Nb})^{0.8}(p_{O_2})} \quad [7]$$

$$\Delta G_2 = -RT\ln K_2 = -RT\ln\frac{(a_{SiO_2})}{(a_{Si})(p_{O_2})} \quad [8]$$

$$\Delta G_3 = -RT\ln K_3 = -RT\ln\frac{(a_{MoO_2})}{(a_{Mo})(p_{O_2})} \quad [9]$$

Because the activity of pure substances is unity, it is appropriate to consider $a_{Nb_2O_5} = a_{SiO_2} = a_{MoO_2} = 1$. Assuming the oxygen partial pressure in dry air to be 0.21, the minimum activities of Nb, Si, and Mo to form the scales of Nb_2O_5, SiO_2, and MoO_2, respectively, have been calculated for the temperatures of interest using Eqs. [7] through [9], and are presented in Table IV. The results in Table IV indicate the following: (a) the minimum activity of Si for the formation of SiO_2 is almost one order of magnitude less than that of Nb for the formation of Nb_2O_5, and (b) the minimum activity of Mo required for the formation of MoO_2 is significantly higher than those of Nb and Si for their respective oxides by several orders of magnitude. Because the activity of Nb in the ternary Nb-Si-Mo alloys is considerably higher than those of Si and Mo, the oxide scales examined in this study exhibit considerably larger volume fraction of Nb_2O_5, compared to that of SiO_2. Hence, a lower volume fraction of SiO_2 is present in the oxide scales, in spite of its lower free energy of formation, compared to those of the other oxides (shown in Table IV). The probability of presence of the MoO_2 at a given location of the oxide scale is less, because of its higher free energy of formation, relatively large minimum activity of Mo (as shown in Table IV) required for the reaction (Eq. [6]), and the relatively lower atomic concentration of Mo (compared to that of Nb) present in the investigated ternary Nb-Si-Mo alloys. Thus, only a small weight fraction of Mo could be detected in the oxide scales of the high Mo containing ternary alloys through EDX analysis. It may also be noted that in the alloys B and D, the concentration of Mo exceeds that of Si.

B. *Oxidation Kinetics*

The data representing the variation of weight gain per unit area (ΔW) of the sample due to oxidation with time (t) have been analyzed to determine the kinetic parameters, namely, the oxidation exponent (n), the general rate constant (k), and the parabolic rate constant (k_p). The gain in mass, ΔW, is related to time, t, through a power-law equation:

$$(\Delta W)^n = kt \quad [10]$$

Equation [10] implies that

$$\ln(\Delta W) = (1/n)\ln k + (1/n)\ln t \quad [11]$$

Hence, the values of n can be obtained by determination of the slope of the best fit line for the plot

showing the variation of the logarithm of ΔW with the logarithm of t, and taking its reciprocal. The relationship between the average rate of mass gain with time in 24 hours of exposure is parabolic for $n = 2$ and linear for $n = 1$. If parabolic rate law is assumed for the investigated Nb-Si-Mo alloys, k_p can be determined from the following relation:

$$(\Delta W)^2 = k_p \cdot t + C \qquad [12]$$

Thus, the value of k_p can be obtained graphically from the slope of the best-fit line for the plot showing the variation of $(\Delta W)^2$ with t. A higher value of k_p would imply a larger rate of oxidation at a given temperature. Typical plots showing the oxidation kinetics in terms of ln (ΔW) against ln (t), as well as $(\Delta W)^2$ against t, for the selected alloys are shown in Figures 14 and 15, respectively.

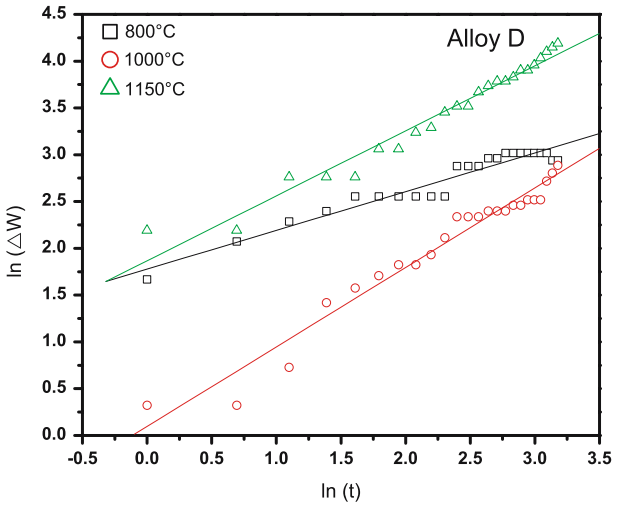


Fig. 14—Plots showing the variation of ln (ΔW) against ln (t) for the alloy D exposed to 800 °C, 1000 °C, and 1150 °C for 24 h.

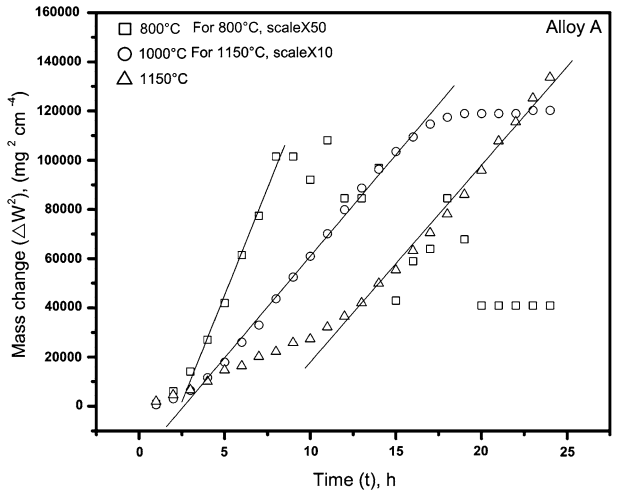


Fig. 15—Plots showing the variation of $(\Delta W)^2$ against (t) for the alloy A exposed to 800 °C, 1000 °C, and 1150 °C for 24 h.

The results in Figure 14 indicate that the data in the logarithmic plot following Eq. [11], in general, exhibit considerable scatter to limit any reliable estimate of the oxidation exponent (n) and the general rate constant (k) for the investigated alloys, if the entire period of 24 hours is considered. However, in a few cases, such as oxidation of the alloys A and C at 1150 °C and that of alloy B at 800 °C, the linear fit through regression analysis of the data for the plot of ln (ΔW) with ln (t) appears to be appropriate, considering that values of the linear regression coefficient R^2 exceed 0.99. The magnitudes of n and k only for these data (oxidation data of the alloys A and C at 1150 °C and that of alloy B at 800 °C) have been estimated and are shown in Table V. Because the values of n (as shown in Table V) determined for the time span of 1 to 24 hours of exposure lie between 1 and 2, it is inferred that the oxidation kinetics of the alloys A and C at 1150 °C, and that of alloy B at 800 °C, follows a rate-law intermediate between linear and parabolic. Analysis of the oxidation kinetics prevalent during the entire period of 24 hours using Eqs. [10] and [11] is not appropriate for the alloy B at 1000 °C and 1150 °C, and for the alloys A and C at temperatures other than 1150 °C.

The values of k_p are shown in Table VI. The results in Figure 15 represent, in general, three different regimes of the rate of oxidation. The first stage is characterized by a slow rate of oxidation. The slower rate of oxidation in the initial stage can be attributed to the formation of an adherent and continuous scale by nucleation and growth of the oxides. The initial rate of oxidation could also be slow due to either an incubation period or an activation barrier to nucleation and growth. In the second stage, the oxidation rate in terms of mass gain is much faster than that in the first stage, and the square of mass gain (ΔW) is almost linear with time, suggesting a parabolic rate law. The parabolic rate law is attributed to steady growth of the oxide scale, controlled by diffusion of ions involved in oxidation. As the oxide scale reaches a critical thickness, the third regime of irregular changes in mass is initiated. The third stage of oxidation displays different possibilities: the mass gain either remains constant, drops, or increases gradually with exposure time during oxidation. The increase in the oxide layer thickness in the third regime is accompanied by a sharp increase in the biaxial compressive stress due to volume expansion owing to the formation of the SiO_2 and Nb_2O_5, as discussed in Section I. The Pilling–Bedworth[41] ratios of SiO_2 and Nb_2O_5 are 2.15 and 2.68, respectively.[42] As a result of the compressive stress, the thin, brittle, and loosely bonded oxide scale is subjected to buckling, followed by cracking.[42] Cracking of the oxide scale exposes more surface area for oxidation and allows the penetration of oxygen to the alloy beneath. However, if cracking is accompanied by spallation, a reduction in mass is the natural consequence. Hence, it is proposed that the surface cracks may lead to either high mass gain due to enhanced rate of oxidation or a reduction in mass caused by spallation. As a result, severe randomness is observed in the plots representing the variation of ln (ΔW) with ln (t) for some of the investigated

Table V. Values of Oxidation Exponent, n, and General Rate Constant, k, Obtained for Oxidation of Selected Alloys during Exposure at Different Temperatures for 24 Hours

Alloy	Temperature (°C)	Total Time Span (h)	n	k ($mg^n cm^{-2n} h^{-1}$)	k ($mg^n cm^{-2n} s^{-1}$)	*R-sq*
A	1150	1 to 24	1.50	45.11242	1.25×10^{-2}	0.993
B	800	1 to 24	1.08	8.041068	2.2×10^{-3}	0.997
C	1150	1 to 24	1.73	61.7624	1.71×10^{-2}	0.992

Table VI. Values of Parabolic Rate Constant, k_p, Corresponding to Exposure at 800 °C, 1000 °C, and 1150 °C for Alloys A, B, C, and D

Alloy	Temperature (°C)	Time Span (h)	k_p ($mg^2\ cm^{-4}\ h^{-1}$)	k_p ($mg^2\ cm^{-4}\ s^{-1}$)	*R-sq*
A	800	3 to 8	319.4	8.9×10^{-2}	0.980
	1000	3 to 17	8248.8	2.29	0.996
	1150	11 to 24	800.3	2.2×10^{-1}	0.991
B	800	6 to 24	746.7	2.1×10^{-1}	0.997
	1000	1 to 22	94.9	2.6×10^{-2}	0.985
	1150	9 to 19	194.0	5.4×10^{-2}	0.983
C	800	1 to 21	4.4	1.2×10^{-3}	0.976
	1000	1 to 12	9098.3	2.52	0.997
	1150	1 to 24	203.1	5.6×10^{-2}	0.992
D	800	1 to 16	24.3	6.7×10^{-3}	0.920
	1000	3 to 19	9.6	2.7×10^{-3}	0.955
	1150	5 to 19	166.0	4.6×10^{-2}	0.985

alloys, particularly when the entire period of 24 hours is considered.

Changes in the oxidation kinetic parameters with the time of exposure due to alterations in the operating mechanisms have been earlier reported for isothermal oxidation of the Ti-6Al-4V alloys by Du *et al.*[43] However in the present study, due to large deviations from a single straight line, k_p has not been calculated for the first and the third stages. Examination of the plots in Figure 15 clearly indicates that the linear variation of $(\Delta W)^2$ against t for the second stage dominates the oxidation kinetics for all the investigated ternary alloys. Thus, the data corresponding to the second regime have been considered for calculating k_p. The data limits for drawing the best-fit line have been selected such that the regression analyses yield R^2 values greater than 95 pct. Thus, the parabolic rate law could be applied to describe the oxidation kinetics of the alloys only during an intermediate period within the total period of exposure (24 hours). Furthermore, the values of n determined from the slopes of best fit lines in the plots of ln (ΔW) against ln (t) representing the intermediate period, for which k_p has been reported (Table VI), have been found as close to 2. Thus, it is appropriate to use the parabolic rate constant to quantify the oxidation kinetics in the regime, at which $n \approx 2$.

The values of k_p and the few estimated values of n and k agree with those reported for the multicomponent Nb-Si alloys by Menon *et al.*[22] For $MoSi_2$, which has excellent oxidation resistance, the values of n and the order of k_p are found reported as 2.37 and 10^{-8} (in $mg^2\ cm^{-4}\ s^{-1}$), respectively, for the isothermal exposure at 1200 °C.[44] The orders of magnitude of k_p values found for isothermal oxidation of the Nb-Si-Mo alloys in the present study are considerably higher than those of $MoSi_2$, suggesting much faster kinetics of oxidation in the former. The estimated values of k_p exhibited by the alloys B and A for isothermal oxidation at 800 °C and 1000 °C, respectively, are the highest among all the investigated alloys. The higher values of the rate constants are consistent with the observation of the highest mass gain in the alloys B and A on isothermal exposure at 800 ° (Figure 5(a)) and 1000 °C (Figure 5(b)), respectively, for 24 hours.

C. *Oxidation Mechanisms*

Pure niobium[32] forms an amorphous scale of Nb_2O_5 at temperatures less than 450 °C, acting as a diffusion barrier to oxygen anions. It has been proposed[32] that at temperatures less than 600 °C, atomic oxygen is first adsorbed on the outer surface layers of the scale, undergoes dissociation, following which the anions diffuse into the scale. The initial oxide scale breaks down in the range of 450 °C to 600 °C, as the amorphous Nb_2O_5 gets transformed to crystalline α-Nb_2O_5. The α-Nb_2O_5 transforms to β-Nb_2O_5 at around 850 °C. If Mo is present in solid solution, β-Nb_2O_5 is formed at a lower temperature. In another study on the multicomponent Nb-Si–based alloy, Menon *et al.*[22] have found distinct evidence of oxygen diffusion and enrichment in the body centered cubic structured, Nb-rich β phase, and not inside the Nb_5Si_3. The oxidation of the multicomponent alloys involves oxygen supersaturation of the β phase, followed by precipitation of oxides of the alloying elements.

Nonisothermal thermogravimetric studies in the present work have shown initiation of oxidation in the

Mo-rich alloys at a lower temperature, regardless of the Si concentration. At temperatures above 550 °C, the increased rate of diffusion of oxygen from the scale into the alloy causes anion deficiency in the scale. Studies by Sims *et al.*[45] have shown that the rate of diffusion of oxygen is slightly higher in the Nb-Mo alloys, compared to that in Nb. In addition, the solubility of oxygen in the bcc Nb_{ss} phase is high and it increases with temperature.[34] A previous study[35] has shown that the presence of Mo increases the volume fraction of both primary and secondary Nb_{ss} in the microstructure, which would further increase the concentration of dissolved oxygen. In addition, Floquet *et al.*[46] have shown that the metallic Mo begins to oxidize on heating above 350 °C forming its thermodynamically most stable oxide, MoO_3. Hence, besides the diffusivity and solubility of oxygen in the alloys having higher Mo concentration, the tendency of Mo to oxidize is expected to trigger the oxidation process at a slightly lower temperature during the nonisothermal heating.

It is interesting to note that the isothermal oxidation experiments at 800 °C have shown higher mass gain in the hypereutectic alloys having the $(Nb,Mo)_5Si_3$ as the primary phase, irrespective of their Mo content. The higher oxidation rate of the hypereutectic alloys at 800 °C could be attributed to the tendency of the $(Nb,Mo)_5Si_3$ phase to undergo accelerated oxidation in the temperature range of 700 °C to 1000 °C, as reported previously.[17] In addition, the alloys A and B with a significantly larger volume fraction of the eutectic also have a high area fraction of the Nb_{ss}-$(Nb,Mo)_5Si_3$ interfaces, which probably provide short circuit paths for diffusion of oxygen. In an earlier investigation on the oxidation behavior of the multicomponent Nb-Si–based alloys, Menon *et al.*[22] have shown that the grain boundaries and interfaces enhance the rate of diffusion of oxygen. At lower temperatures, the diffusion through grain boundaries or interfaces is expected to be much faster than lattice diffusion.

The Nb_2O_5 and SiO_2 form a eutectic with melting point at 1449 °C,[47] while the melting point of Nb_2O_5 is roughly 1550 °C. Even if the temperatures of isothermal oxidation experiments are lower in this study, it is likely that the diffusivity of oxygen would be higher in the eutectic mixture. Hence, alloying of Nb with Si is detrimental for oxidation resistance.

The alloys with relatively lower concentrations of Mo have shown higher mass gain at 1000 °C and 1150 °C, irrespective of their hypo- or hypereutectic character. Because the sublimation temperature of MoO_3 is 704 °C, loss of Mo through volatilization in the initial stages of oxidation is possible. Although mass gain could be partially counterbalanced by the loss of MoO_3 in the alloys containing high concentrations of Mo, the larger mass gain by the alloy B compared to that by A, in spite of containing much higher Mo content in the former, implies that loss of Mo is not significant in this alloy. The beneficial effect of Mo at 1000 °C and 1150 °C is obvious, when one compares the nonisothermal (Figures 2 through 4) and isothermal oxidation behavior (Figure 5) of the hypoeutectic Nb-Si-Mo alloys with that of Nb and Nb-10Si alloy. The formation of MoO_2 or MoO_3 is suppressed by that of Nb_2O_5 and SiO_2, which is explained on the basis of thermodynamic parameters in Section A.

The Mo content of Nb_{ss} is much higher, compared to that of α-$(Nb,Mo)_5Si_3$ (Table I). The results of this study agree well with that of a few previous publications,[16,32–33] which have suggested that alloying of Nb with Mo is beneficial for oxidation resistance. MoO_2 has the free energy of formation closest to that of Nb_2O_5,[37] among the oxides of Mo. Also the radii of Mo^{4+} and Nb^{5+} ions are 0.70 and 0.69 Å, respectively, whereas Mo^{6+} ion has a radius of 0.62 Å. Hence, it is intuitive that Mo dissolves in the Nb_2O_5 in the form of Mo^{4+} substituting the Nb^{5+} sites. Both Nb_2O_5 and SiO_2 are *n*-type semiconductors containing excess of positively charged anion vacancies and electrons.[48] A higher concentration of anion vacancies is expected to increase the rate of diffusion of oxygen anions from the oxide-air interface to metal-oxide interface. In contrast to the *n*-type character of Nb_2O_5 and SiO_2, MoO_2 is an amphoteric conductor. Because the Mo^{4+} cations have a lower valence compared to that of Nb^{5+}, substitution of the former in the Nb_2O_5 is expected to lower the concentration of anion vacancies and reduce the rate of oxygen anion diffusion through the oxide scale.

Rapid surface diffusion in porous scale of Nb_2O_5 is the cause of its unprotective character. The oxides of Mo have lower melting points compared to that of Nb_2O_5, as a result of which the sinterability of the oxide scale is improved significantly. Reduction of void surface area through sintering of the oxide scale is expected to reduce the rate of oxygen penetration through surface diffusion significantly. It is possible that after an initial loss, the improvement of sinterability of the oxide scale enhances its protective capability, and arrests the change in mass.[32] The improved sinterability of the oxide scale explains why the top surfaces of the oxide scales in alloy D at 800 °C and 1000 °C are relatively smooth, compared to that of the hypereutectic alloys. On the other hand, the sinterability of the oxide scales of the low Mo-containing alloys, A and C is also expected to improve with increasing temperature, causing a reduction in the porosity content. The improved sinterability could explain the greater smoothness of the oxide scales formed at 1150 °C than that at 1000 °C. Reduced porosity and greater smoothness of the oxide scale surface is expected to significantly decrease the level of penetration of oxygen. Hence, the lower mass gain observed in the alloys A and C at 1150 °C than that at 1000 °C is attributed to the improved sinterability, reduced porosity, and greater smoothness of their respective oxide scales.

Examination using EDX analysis of cross section of the oxide scales formed at 1150 °C on the high Mo containing alloys B and D has shown the presence of a layer of SiO_2 at the oxide-metal interfaces, which is capable of partially protecting against further oxidation by acting as a diffusion barrier for penetration of oxygen. The SiO_2 forms as the inner layer, because of its slower growth rate than that of the oxides of Nb. The slow growth rate of SiO_2 is responsible for its lower parabolic rate constant,[3,49] compared to that of other

oxides. Observation of oxygen enrichment to a depth of 50 μm beneath the SiO_2 layer in the alloys B and D indicates that the oxidation rate is controlled by diffusion of oxygen anions across the amorphous SiO_2. The formation of SiO_2 layer is observed only at the alloy-oxide interface of high Mo-containing alloys, and not in those having lower concentration. The Mo substitution of Nb sites results in increasing the activity of Si in comparison to that of Nb. Because Si is more reactive than Nb and Mo, and SiO_2 is thermodynamically more stable, it is able to form a continuous layer in oxide scales of the alloys B and D. In the temperature range of 800 °C to 1150 °C, the diffusivities of oxygen anion through the crystalline Nb_2O_5 and amorphous SiO_2 are 4.7×10^{-3} and 1.5×10^{-2} cm^2/s, respectively, while the activation energies are 277 and 71 kCal/g atom, respectively.[48] In spite of slightly higher diffusivity, the amorphous SiO_2 scale is more protective because of its adherent and continuous character, unlike that of Nb_2O_5, which is porous and discontinuous.

V. CONCLUSIONS

The metallic Nb, hypoeutectic Nb-10Si alloy, and the ternary hypereutectic Nb-19Si-5Mo (alloy A) and Nb-18Si-26Mo (alloy B), and hypoeutectic Nb-13Si-4Mo (alloy C) and Nb-12Si-15Mo (alloy D) alloys, all prepared by arc melting, have been subjected to nonisothermal oxidation in the temperature range of 50 °C to 1300 °C, and isothermal oxidation at 800 °C, 1000 °C, and 1150 °C in dry air. The most important conclusion in this study is that the hypo- or hypereutectic character of the alloys defines the oxidation behavior at 800 °C, while the effect of Mo content appears to be predominant at temperatures of 1000 °C to 1150 °C and under nonisothermal conditions. The other important observations in this work are summarized as follows.

1. Nonisothermal thermogravimetric analyses have shown the highest mass gain in case of the Nb-10Si alloy, followed by that of Nb, and the alloys A, C, B, and D. Among the Nb-Si-Mo alloys, those with a lower concentration of Mo have shown a higher gain in mass, compared to those having a higher Mo content.
2. Thermogravimetric experiments at a given heating rate shows initiation of mass gain at a lower temperature for the high Mo-containing alloys, B and D, compared to that for the low Mo-containing alloys, A and B.
3. The nonisothermal oxidation behavior depends on the heating rates. The gain in mass tends to be higher for slower heating rates. On the other hand, oxidation begins at higher temperature as the heating rate increases.
4. The results of differential thermal analysis indicate that the oxidation reaction involving the Nb_{ss} phase precedes that of α-$(Nb,Mo)_5Si_3$. The temperature of the oxidation reactions depends on the concentration of substitutional solute, Mo in either Nb_{ss} or in the α-$(Nb,Mo)_5Si_3$.
5. Isothermal oxidation studies have shown that none of the ternary alloys experiences catastrophic oxidation in the temperature range of 800 °C to 1150 °C within the 24-hour period of exposure.
6. The highest mass gain at 800 °C has been recorded in the alloy B, and those at 1000 °C and 1150 °C in the alloy A. At 800 °C, the hypereutectic alloys A and B with primary α-$(Nb,Mo)_5Si_3$ in the microstructure are more prone to oxidation, irrespective of their Mo concentration. At 1000 °C or 1150 °C, the alloys with lower Mo content show larger mass gain, regardless of which side of the eutectic the bulk composition lies.
7. The oxide scale is made of Nb_2O_5 and SiO_2, with the volume fraction of the former being much higher.
8. In general at any temperature, the oxide scales on the alloys showing relatively large mass gain have been found to be porous, while those accompanying oxidation with lower mass gain appear to be smooth. Alloying with Mo appears to improve the sinterability of the oxide scale.
9. The alloys B and D with high Mo content have shown a continuous layer of amorphous SiO_2 formed at the alloy-oxide interface on exposure at 1000 °C and 1150 °C, which probably improves the oxidation resistance.
10. For all the investigated alloys and temperatures in the range of 800 °C to 1150 °C, the kinetic data from an intermediate, but the dominant time span within the 24 hours of exposure, appears to fit in the parabolic rate law. The kinetic parameters are directly related to the extent of oxidation leading to mass gain, and are a function of time and temperature of exposure, besides the alloy composition.

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REFERENCES

1. P.R. Subramanian, T.A. Parthsarathy, M.G. Mendiratta, and D.M. Dimiduk: *Scripta Mater.*, 1995, vol. 32, pp. 1227–32.
2. B.P. Bewlay, M.R. Jackson, and H.A. Lipsitt: *Metall. Mater. Trans. A*, 1996, vol. 27A, pp. 3801–08.

3. R. Mitra: *Int. Mater. Rev.*, 2006, vol. 51 (1), pp. 13–64.
4. M.G. Mendiratta, J.J. Lewandowski, and D.M. Dimiduk: *Metall. Mater. Trans. A*, 1991, vol. 22A, pp. 1573–81.
5. M.G. Mendiratta and D.M. Dimiduk: *Metall. Mater Trans. A*, 1993, vol. 24A, pp. 501–04.
6. D.M. Shah, D.L. Anton, D.P. Pope, and S. Chin: *Mater. Sci. Eng. A*, 1995, vol. A192, pp. 658–75.
7. W.Y. Kim, H. Tanaka, A. Kasama, R. Tanaka, and S. Hanada: *Intermetallics*, 2001, vol. 9, pp. 521–27.
8. W.Y. Kim, H. Tanaka, A. Kasama, and S. Hanada: *Intermetallics*, 2001, vol. 9, pp. 827–34.
9. J. Sha, H. Hirai, T. Tabaru, A. Kitahara, H. Ueno S. Hanada: *Metall. Mater. Trans. A*, 2003, vol. 34A, pp. 85–94.
10. J.H. Kim, T. Tabaru, and H. Hirari: *Mater. Trans., JIM*, 2003, vol. 44, pp. 673–76.
11. W.Y. Kim, H. Tanaka, M.S. Kim, and S. Hanada: *Mater. Sci. Eng., A*, 2003, vol. A346, pp. 151–58.
12. J. Shah, H. Hirari, T. Tabaru, A. Kitahara, H. Ueno S. Hanada: *Metall. Mater. Trans. A*, 2003, vol. A34, pp. 2861–71.
13. J.H. Kim, T. Tabaru, H. Hirari, A. Kitahara, and S. Hanada: *Scripta Mater.*, 2003, vol. 48, pp. 1439–44.
14. J. Sha, H. Hirai, T. Tabaru, A. Kitahara, H. Ueno S. Hanada: *Mater. Sci. Eng., A*, 2003, vol. A364, pp. 151–58.
15. K. Chattopadhyay, G. Balachandran, R. Mitra, and K.K. Ray: *Intermetallics*, 2006, vol. 14, pp. 1452–60.
16. O. Kubaschewski and B.E. Hopkins: *J. Less Comm. Met.*, 1960, vol. 2, pp. 172–80.
17. M.R. Jackson, R.G. Rowe, and D.W. Skelly: *Mater. Res. Soc. Symp. Proc.*, 1995, vol. 364, pp. 1339–44.
18. D.L. Douglass: *J. Less Comm. Met.*, 1963, vol. 5, pp. 151–57.
19. P.R. Subramanian, M.G. Mendiratta, and D.M. Dimiduk: *Mater. Res. Soc. Symp. Proc.*, 1994, vol. 322, pp. 491–02.
20. P.R. Subramanian, M.G. Mendiratta, D.M. Dimiduk, and M.A. Stucke: *Mater. Sci. Eng., A*, 1997, vols. A239–A240, pp. 1–13.
21. P.R. Subramanian, M.G. Mendiratta, and D.M. Dimiduk: *JOM*, 1996, vol. 48, pp. 33–38.
22. E. Sarath, K. Menon, M.G. Mendiratta, and D.M. Dimiduk: in *Structural Intermetallics*, K.J. Hemker, D.M. Dimiduk, H. Clemens, R. Darolia, H. Inui, J.M. Larse, V.K. Sikka, M. Thomas J.D. Whittenberger, eds., TMS, Warrendale, PA, 2001, pp. 591–600.
23. J. Geng, P. Tsakiropoulos, and G. Shao: *Intermetallics*, 2006, vol. 14, pp. 227–35.
24. K.S. Chan: *Mater. Sci. Eng. A*, 2002, vols. A329–A331, pp. 513–22.
25. V. Behrani, J.A. Thom, M.J. Kramer, and M. Akinc: *Intermetallics*, 2006, vol. 14, pp. 24–32.
26. D.L. Davinson and K.S. Chan: *Metall. Mater. Trans. A*, 1999, vol. 30A, pp. 2007–18.
27. D.L. Davinson, K.S. Chan, and D.L. Anton: *Metall. Mater. Trans. A*, 1996, vol. 27A, pp. 3007–18.
28. K. Zelenitsas and P. Tsakiropoulous: *Mater. Sci. Eng., A*, 2006, vol. 416, pp. 269–80.
29. Y. Murayama and S. Hanada: *Sci. Technol. Adv. Mater.*, 2002, vol. 3, pp. 145–56.
30. K.S. Chan: *Metall. Mater. Trans. A*, 2004, vol. 35A, pp. 589–98.
31. K.S. Chan: *Oxid. Met.*, 2004, vol. 61 (3–4), pp. 165–94.
32. B.B. Argent and B. Phelps: *J. Less Common Met.*, 1996, vol. 2, pp. 181–90.
33. G.L. Miller and F.G. Cox: *J. Less Common Met.*, 1960, vol. 2, pp. 207–22.
34. S.J. Balsone, B.P. Bewlay, M.R. Jackson, P.R. Subramanian, J.-C. Zhao, A. Chatterjee, and T.M. Heffernan: Report No. 2001CRD098, GE Research and Development, GE Power Systems, Schenectady, NY, 2001.
35. K. Chattopadhyay, R. Mitra, and K.K. Ray: *Mater. Sci. Eng., A.*, 2007, vol. A456, pp. 358–63.
36. M.E. Schlesinger, H. Okamoto, A.B. Gokhale, and R. Abbaschian: *J. Phase Equilib.*, 1993, vol. 14, pp. 502–09.
37. I. Barin: *Thermochemical Data of Pure Substances*, VCH, Weinheim, Germany, 1989, vols. 1 and 2, pp. 924, 931, 1033, 1359.
38. M.S. Samant, A.S. Kerkar, S.R. Bharadwaj, and S.R. Dharwadkar: *J. Alloys Compd.*, 1992, vol. 187, pp. 373–79.
39. T.A. Parthasarathy, M.G. Mendiratta, and D.M. Dimikuk: *Acta Mater.*, 2002, vol. 50 (7), pp. 1857–68.
40. S. Paswan, R. Mitra, and S.K. Roy: *Intermetallics*, 2007, vol. 15, pp. 1217–27.
41. N.B. Pilling and R.E. Bedworth: *J. Inst. Met.*, 1923, vol. 29, pp. 529–82.
42. U.K. Chatterjee, S.K. Bose, and S.K. Roy: *Environmental Degradation of Metals*, Marcel Dekkar, Inc., New York, NY, pp. 283–411.
43. H.L. Du, P.K. Datta, D.B. Lewis, and J.S. Burnell-Gray: *Corros. Sci.*, 1994, vol. 36 (4), pp. 631–42.
44. R. Mitra and V.V. Rama Rao: *Mater. Sci. Eng., A.*, 1999, vol. A260, pp. 146–60.
45. C.T. Sims, W.D. Kloff, and R.I. Jaffee: *Trans. ASM*, 1957, vol. 51, pp. 256–81.
46. N. Floquet, O. Bertrand, and J.J. Heizmann: *Oxid. Met.*, 1992, vol. 37, pp. 253–80.
47. E.M. Levin, C.R. Robbins, and H.F. McMurdle: *Phase Diagram for Ceramics*, The American Ceramic Society, Westerville, Ohio, 1964, pp. 142–363.
48. O. Kubaschewski and B.E. Hopkins: *Oxidation of Metals and Alloys*, Butterworth and Co, London, 1967, pp. 1–69.
49. S.V. Raj: *Mater. Sci. Eng., A.*, 1995, vol. A201, pp. 229–41.